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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

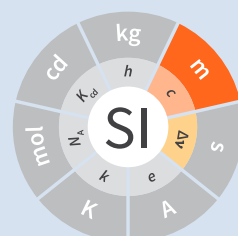
Part 2

Strontium 88 Ion ()

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CCL-CCTF Frequency Standards Working Group

April 2018
avril 2018



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Strontium 88 Ion ($^{88}\text{Sr}^+$)**

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1. Recommended value [1] of the frequency in the CIPM List of Frequencies

$$f(^{88}\text{Sr}^+) = 444\,779\,044\,095\,486.5\text{ Hz}$$

equivalent to

$$\lambda(^{88}\text{Sr}^+) = 674\,025\,590.863\,134\text{ fm},$$

with a relative standard uncertainty of 1.5×10^{-15} .

This radiation was endorsed by the CIPM as a secondary representation of the definition of the second [2].

2. Source data

Adopted value $f(^{88}\text{Sr}^+) = 444\,779\,044\,095\,486.5\text{ Hz}$

With relative standard uncertainty: 1.5×10^{-15}

calculated from

$f(^{88}\text{Sr}^+)/\text{Hz}$	u/Hz	source data
444 779 044 095 484.6	1.5	[3]
444 779 044 095 484	15	[4]
444 779 044 095 485.5	0.9	[5]
444 779 044 095 486.71	0.24	[6] ^(a)
444 779 044 095 485.27	0.75	[7]

(a) Since the result is largely dominated by the single measurement of ref. [6], the final uncertainty has been increased by a factor of three following the procedures outlined in ref. [8].

as a weighted mean of the above values.

$^{88}\text{Sr}^+$, $5s\ ^2\text{S}_{1/2} - 4d\ ^2\text{D}_{5/2}$ unperturbed optical transition

References

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